



Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard



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1. Executive Summary

This report documents the complete Mutual Fund Analytics Platform developed during a 7-day Data Analyst Internship at Bluestock Fintech Pvt. Ltd. (June 2026). The project ingests publicly available mutual fund data from AMFI India and mfapi.in, transforms it through a robust Python-based ETL pipeline, stores it in a normalised SQLite database, and presents actionable insights through an interactive 4-page Power BI dashboard.

The platform covers 40 real mutual fund schemes across 10 fund houses, analysing 46,000+ daily NAV records and 32,778 investor transactions spanning January 2022 to May 2026. Key risk-adjusted metrics including Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Value at Risk (VaR) were computed for all 40 schemes.

46,000+	32,778	40	10	Rs.81L Cr	Rs.31K Cr
NAV Records	Transactions	Fund Schemes	Fund Houses	Industry AUM	SIP All-Time High

Key Outcomes:

- Built fully automated ETL pipeline fetching live NAV from mfapi.in REST API
- Designed 5-table SQLite star schema with 46,000+ fact records loaded successfully
- Generated 9 publication-quality EDA charts with documented insights
- Computed 7 risk-adjusted performance metrics for all 40 fund schemes
- Developed 4-page interactive Power BI dashboard with drill-through and slicers
- Built fund recommender engine and VaR/CVaR risk model for advanced analytics
- Documented findings in 15–20 page report with 12-slide presentation deck

2. Problem Statement

Despite India’s mutual fund industry managing over Rs.81 lakh crore in AUM (Dec 2025) across 26.12 crore investor folios, individual investors and financial advisors face significant challenges in making data-driven fund selection decisions. This project addresses five core business problems:

ID	Problem	Solution Implemented
P1	Data Fragmentation	Unified ETL pipeline consolidating all data into single SQLite database
P2	Performance Comparison Gap	Computed Sharpe, Sortino, Alpha, Beta, Max Drawdown for all 40 schemes
P3	No Benchmark Tracking	Joined NAV data with Nifty 50/100 benchmark, computed tracking error
P4	Investor Behaviour Blind Spot	Analysed 32,778 transactions by state, age group, city tier, transaction type
P5	Slow Static Reporting	Built live Power BI dashboard refreshable directly from ETL pipeline

3. Data Sources & Datasets

All datasets are derived from publicly available, freely accessible sources including AMFI India (amfiindia.com), mfapi.in REST API, NSE India, and BSE India. No proprietary or confidential data was used in this project.

Dataset Inventory

#	File	Rows	Description	Source
01	fund_master.csv	40	40 real MF schemes — AMFI codes, expense ratio, risk grade	AMFI India
02	nav_history.csv	46,000	Daily NAV Jan 2022–May 2026 for all 40 schemes	mfapi.in
03	aum_by_fund_house.	~90	Quarterly AUM for 10 fund houses 2022–2025	AMFI Quarterly
04	monthly_sip_inflows.csv	4v8	Month-wise SIP inflow, accounts, AUM — Jan 2022–Dec 2025	AMFI Monthly
05	category_inflows.csv	~144	Net inflows by fund category for FY 2024-25	AMFI India
06	industry_folio_count.csv	2s1v	Total MF folios by type — Equity, Debt, Hybrid	AMFI India
07	scheme_performance	40	1yr/3yr/5yr returns, Sharpe, Sortino, Alpha, Beta	Computed
08	investor_transactions.csv	32,778	SIP + Lumpsum + Redemption for 5,000 investors	Simulated
09	portfolio_holdings	~320	Top equity holdings per fund as of Dec 2025	AMFI India
10	benchmark_indices.csv	8,000	Daily Nifty 50, Nifty 100, BSE SmallCap values	

4. System

Architecture & ETL Pipeline

NSE/BSE

The project follows a classic 5-layer data engineering architecture — Extract, Transform, Load, Analyse, Visualise — mirroring real-world fintech data pipelines used at companies like Zerodha, Groww, and Paytm Money.

Layer Name		Tools	Output
L1	Extract	requests, mfapi.in REST API, AMFI CSV	10 raw CSV files + live NAV JSON
L2	Transform	Python, Pandas — clean, re-shape, forward-fill	10 cleaned CSVs in data/processed/
L3	Load	SQLite, SQLAlchemy — star schema, indexes	bluestock_mf.db (5 tables, 80K+ rows)
L4	Analyse	Jupyter, NumPy, SciPy — metrics, EDA	9 charts, 7 metrics, 5 advanced models
L5	Visualise	Power BI Desktop — 4 pages, slicers, drill-through	bluestock_mf_dashboard.pbix

Database Star Schema

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category, expense_ratio	40
dim_date	Dimension	date_id (PK), date, year, month, quarter	1,500
fact_nav	Fact	amfi_code (FK), date (FK), nav, daily_return	46,000
fact_transactions	Fact	tx_id (PK), investor_id, amfi_code, amount, type	32,778
fact_performance	Fact	amfi_code (FK), sharpe, alpha, beta, max_drawdown	40

5. Exploratory Data Analysis (EDA)

Day 3 EDA analysis produced 9 publication-quality charts using Plotly, Seaborn, and Matplotlib. Analysis covered NAV trends, AUM growth, SIP inflows, investor demographics, geographic distribution, folio growth, return correlations, and sector allocation.

#	Chart Title	Key Insight
1	NAV Trend — All 40 Schemes (2022–2026)	All 40 NAVs grew 2x–3x. 2023 bull run drove 15–22% gains. Oct–Nov 2024 saw 7% correction.
2	AUM Growth by Fund House (2022–2025)	SBI MF leads at Rs.12.5L Cr, nearly doubling from Rs.6.5L Cr in 2022.
3	Monthly SIP Inflow Trend (2022–2025)	SIP inflows tripled from Rs.11,000 Cr to Rs.31,002 Cr — structural growth, not cyclical.
4	Category Inflow Heatmap	Large Cap dominates year-round. ELSS spikes Jan–Mar (tax-saving season).
5	Investor Age Demographics	26–35 age group has most investors. 46–55 cohort invests highest median SIP (Rs.12,000/mo).
6	Geographic Distribution	72% of SIP value from T30 cities. Maharashtra alone contributes 20%+.
7	Folio Count Growth	Folios doubled from 13.26 Cr to 26.12 Cr in 4 years — 18.5% CAGR.
8	NAV Return Correlation Matrix	Large Cap funds correlate at 0.85–0.95 — multiple holdings provide minimal diversification.
9	Sector Allocation Donut	BFSI (32%) + IT (18%) = 50% of equity portfolios — concentrated sector exposure.

6. Fund Performance Analytics

Day 4 computed all key risk-adjusted performance metrics from 46,000+ daily NAV records. A composite Fund Scorecard (0–100) was built using weighted ranking across 5 metrics.

Performance Metrics Computed

Metric	Formula	Purpose
Daily Return	$\text{nav}_t / \text{nav}_{t-1} - 1$	Base metric for all calculations
CAGR	$(\text{NAV}_{\text{end}} / \text{NAV}_{\text{start}})^{1/n} - 1$	Annualised return over 1/3/5 years
Sharpe Ratio	$\frac{R_p - R_f}{\sigma_p} \times \sqrt{252}$, $R_f = 6.5\%$	Risk-adjusted return vs RBI rate
Sortino Ratio	$\frac{R_p - \sigma_{\text{down}}}{\sigma_{\text{down}}} \times \sqrt{252}$	Penalises only downside volatility
Alpha	OLS intercept $\times 252$ (vs Nifty 100)	Excess return above benchmark
Beta	OLS slope (fund returns vs Nifty 100)	Market sensitivity (1.0 = market)
/ —		
Max Drawdown	$\min(\text{NAV}_{\text{running_max}} - \text{NAV}_{\text{trough}})$	Worst peak-to-trough loss
Tracking Error	$\text{std}(r_{\text{fund}} - r_{\text{benchmark}}) \times \sqrt{252}$	Deviation from benchmark

Fund Scorecard Weights

Component	Weight	Rationale
3yr CAGR Rank	30%	Primary return metric — most relevant for retail investors
Sharpe Ratio Rank	25%	Risk-adjusted performance — quality of returns
Alpha Rank	20%	Benchmark outperformance — active management value
Expense Ratio	15%	Lower cost = better long-term compounding (inverse rank)
Max Drawdown	10%	Capital protection — lower drawdown = better (inverse rank)

7. Power BI Dashboard

Day 5 produced a 4-page interactive Power BI dashboard with drill-through, slicers, and Bluestock brand theme applied. All pages are cross-filterable and include tooltips.

Page	Title	Visuals	Slicers
1	Industry Overview	4 KPI cards, AUM trend line, AUM by AMC bar, market share donut	Year
2	Fund Performance	Risk vs Return scatter (bubble = AUM), sortable scorecard table, NAV line chart	Fund House, Category, Plan
3	Investor Analytics	SIP by state bar, SIP/Lumpsum/Redemption donut, age group bar, monthly volume line	State, Age Group, City Tier
4	SIP & Market Trends	SIP inflow bar + Nifty 50 line (dual-axis), category heatmap, top 5 categories bar	None

Dashboard KPI Cards



8. Advanced Analytics & Risk Metrics

Day 6 implemented advanced quantitative models including Value at Risk (VaR), Conditional VaR, rolling Sharpe analysis, investor cohort analysis, SIP continuity modelling, fund recommendation engine, and sector concentration (HHI) analysis.

Model	Method	Key Finding
Historical (95%) CVaR (95%)	VaR 5th percentile of daily return distribution Mean of returns below VaR threshold	Small Cap VaR: -2.1% to -2.8% daily vs -1.3% to -1.6% for Large Cap CVaR consistently 1.3–1.5× worse than VaR — fat tail risk in mid/small cap
Rolling Sharpe	90-Day returns.rolling(90).	Sharpe hit 2.0+ during 2023 bull run. Dropped below 0 in Oct–Nov 2024
Investor Analysis	Cohort Group by first transaction year, compute avg SIP	2024 cohort has higher avg SIP than 2022 — rising financial awareness
SIP Continuity	Avg gap between SIP dates, flag > 35 days	~23% of active SIP investors flagged as At-Risk
Fund Recommender	Filter by risk_category, rank by Sharpe	Top 3 funds returned per Low/Moderate/High risk profile
Sector HHI	$\sum w_i^2$ per fund portfolio	Sectoral funds HHI > 0.20 (concentrated). Large Cap HHI < 0.10

9. Key Findings & Recommendations

10 Key Findings

1. SIP Growth Is Structural

[High]

Monthly SIP inflows grew 3× (Rs.11K to Rs.31K Cr) without any sustained reversal, even during market corrections. SIP is now deeply embedded retail savings behaviour.

2. SBI MF Has Unassailable AUM Lead

[High]

SBI MF AUM doubled to Rs.12.5L Cr in 3 years, maintaining 16%+ market share via government-bank distribution network across B30 cities.

3. India MF Participation Doubled

[High]

Folio count grew from 13.26 Cr to 26.12 Cr (18.5% CAGR) driven by digital platforms like Zerodha Coin, Groww, and Paytm Money democratising investing.

4. Large Cap Funds Over-Correlated

[High]

NAV return correlations of 0.85–0.95 among Large Cap funds mean holding SBI Bluechip + HDFC Top 100 + ICICI Bluechip provides almost no diversification benefit.

5. BFSI + IT = 50% of Equity Portfolio Risk

[Medium]

Aggregate sector allocation shows BFSI (32%) and IT (18%) dominate. A banking NPA crisis or IT slowdown would disproportionately impact equity fund returns.

6. T30 Cities Dominate But B30 Growing

[Medium]

72% of SIP value from T30 cities. B30's 28% share is growing as AMFI's higher commission incentive for B30 investors takes effect.

7. Tax-Season ELSS Spike

[Medium]

Category heatmap shows ELSS spikes every Jan–Mar — predictable seasonal demand

driven by 80C tax-saving deadline on March 31.

8. 2023 Bull Run Rewarded SIP Investors

[High]

Investors who maintained SIPs through 2022 bear phase averaged into lower NAVs and captured full 15–22% 2023 recovery.

9. Experienced Investors Invest Most

[Medium]

46–55 age group shows highest median SIP (Rs.12,000/mo) vs Rs.5,000 for 26–35. This segment drives disproportionate AUM despite smaller investor count.

10. 23% SIP Investors At-Risk

[Medium]

Approximately 23% of investors with 6+ SIPs have average gaps > 35 days. These investors need targeted engagement/reminder campaigns.

10. Business Recommendations

For Investors

- Diversify across categories (Large + Mid + Debt) rather than holding multiple Large Cap funds that are 85–95% correlated.
- Maintain SIPs through market corrections — 2022–2023 data shows SIP investors recovered fully and outperformed lump sum investors.
- Consider expense ratio as a key filter — funds under 1% expense ratio (Direct plans) significantly outperform Regular plans over 5+ years.
- Use the Fund Scorecard (composite 0–100) to shortlist funds rather than relying solely on 1-year returns.

For Bluestock Fintech Platform

- Implement automated SIP gap detection — flag investors with 35+ day gaps for re-engagement campaigns.
- Add B30 city-specific fund recommendations leveraging AMFI's higher commission incentive for financial inclusion.
- Build a portfolio overlap checker — show investors correlation between their existing funds to prevent false diversification.
- Schedule daily NAV refresh via cron job (Bonus B1) to keep dashboard data current without manual intervention.
- Expand benchmark tracking to include Nifty Midcap 150 and BSE SmallCap for respective fund categories.

11. Limitations & Future Scope

Current Limitations

- Investor transaction data (08_investor_transactions.csv) is synthetically generated — real transaction patterns may differ.

- SQLite used for development — production deployment would require PostgreSQL with connection pooling.
- Power BI dashboard requires manual refresh — no live streaming data connection implemented.
- NAV history limited to Jan 2022–May 2026 — longer history would improve statistical significance of Sharpe/Beta calculations.
- Fund recommender uses simple Sharpe-based ranking — a production system would incorporate personalisation and risk profiling.

Future Scope (Bonus Features)

- **B1** — Schedule ETL as cron job: auto-fetch NAV from mfapi.in every weekday at 8 PM
- **B2** — Build Streamlit web app as real-time alternative to Power BI dashboard
- **B3** — Monte Carlo simulation projecting NAV growth over 5 years with uncertainty bands
- **B4** — Markowitz Efficient Frontier portfolio optimisation for 5 selected funds
- **B5** — Automated HTML email report generator sending weekly performance summaries

12. Technical Stack & Deliverables

Technology Stack

Category	Tool/Library	Version	Purpose
Language	Python	3.10+	All data processing, ETL, analytics
Data	Pandas	2.0+	DataFrames, cleaning, aggregation
Numerical	NumPy	1.24+	Risk metrics, statistical functions
Statistics	SciPy	1.10+	OLS regression for Alpha/Beta
Viz	Matplotlib	3.7+	Static charts for reports
Viz	Seaborn	0.12+	Heatmaps, distributions
Viz	Plotly	5.x	Interactive charts in Jupyter
Database	SQLite3	Built-in	Local relational database
ORM	SQLAlchemy	2.0	Python-to-SQLite interface
Dashboard	Power BI	Latest	Interactive 4-page dashboard
Version Ctrl	Git + GitHub	Latest	Code versioning, submission
API	mfapi.in	v1	Live NAV data (no auth required)

Final Deliverables Checklist

#	Deliverable	Format	Weight	Status
D1	ETL Pipeline Script	.py	15%	Complete
D2	SQLite Database	.db	10%	Complete
D3	EDA Notebook	.ipynb	15%	Complete
D4	Performance Metrics	.ipynb+CSV	15%	Complete
D5	Interactive Dashboard	.pbix	20%	Complete
D6	Advanced Analytics	.ipynb	10%	Complete
D7	Final Report + Slides	.pdf+.pptx	15%	Complete

13. Conclusion

This 7-day capstone project successfully delivered a full-stack Mutual Fund Analytics Platform that addresses real business problems faced by retail investors and financial advisors in India’s rapidly growing mutual fund industry.

The platform demonstrates end-to-end data engineering competency — from raw data ingestion via REST APIs, through Python-based ETL transformation, SQL database design, statistical analysis, and interactive dashboard development. All 8 project objectives and 7 deliverables were completed within the 7-day timeline.

Key technical achievements include: a normalised SQLite star schema handling 80,000+ rows, 7 risk-adjusted performance metrics computed from scratch using NumPy/SciPy, a composite Fund Scorecard with weighted multi-metric ranking, and a 4-page Power BI dashboard with drill-through interactivity and Bluestock brand theme.

Project Status:
COMPLETE
GitHub Tag: v1.0

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